



BUP CSE FEST 2026 HACKATHON

In association with Poridhi



BANGLADESH UNIVERSITY OF PROFESSIONALS

Mirpur Cantonment, Dhaka - 1216

Preliminary Problem Statement

Smart Campus Energy Optimization Challenge

LLM-Assisted Operator Directive Interpretation

Online Preliminary Round

Item	Details
Round	Online Preliminary
Round Window	7:00 PM - 11:00 PM (4 hours)
Challenge Type	LLM-assisted energy scheduling and optimization
Required Service	Deployed Public HTTP API
Health Endpoint	GET /health
Main Endpoint	POST /optimize-energy
Planning Horizon	24 hourly intervals
Operator Notes	1-3 natural-language notes per scenario
Response Format	Structured JSON

All scenarios and operator notes are synthetic. No live campus, utility, billing, or personal data is required.

00. How to Use This Statement

This document is the canonical specification for the preliminary challenge behavior, API contract, operator-note interpretation, optimization rules, and validation requirements.

IMPORTANT: Deployment, repository, submission, performance, scoring, tie-break, and general participation rules remain defined in the separate Participant Guide & Evaluation Rubric.

Participant document pack

Document	Purpose
Problem Statement (this document)	Defines the scenario, operator-note interpretation, API schema, optimization rules, and validity requirements.
Participant Guide & Evaluation Rubric	Defines deployment, testing, submission, scoring, penalties, and tie-break rules.
Public Sample Cases JSON	Provides worked scenarios for local validation. Public cases are not the hidden judge set.

Inside this statement

Section	Contents
01-03	Scenario, system goal, and end-to-end flow
04-05	Operator directives, clauses, and optimization objective
06-07	API contract and request schema
08	LLM interpretation guardrails
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01. The Scenario

BUP is operating a smart campus using electricity purchased from the grid, rooftop solar generation, and a battery energy storage system. Campus demand, solar availability, and grid tariff vary throughout the day.

The next 24 hours of demand, solar generation, and grid tariff are already provided. In addition, campus operators may send short natural-language notes describing temporary operating conditions that affect the same 24-hour schedule.

Your service must understand the notes, convert relevant instructions into structured directives, apply those directives to the optimization problem, and return a valid low-cost operating plan.

02. What You Are Building

Build one HTTP API service that receives a 24-hour energy scenario plus operator notes and returns both a machine-checkable interpretation of those notes and the final 24-hour energy schedule.

- Interpret every operator note using an LLM or other language-capable generative model.
- Convert relevant notes into one of the supported directive types.
- Mark irrelevant notes as `no_op` instead of inventing an energy rule.
- Validate the interpreted directives before sending them to the optimizer.
- Produce a valid schedule that satisfies the normal GridWise rules and every applicable directive.
- Minimize total grid electricity cost after correctness is satisfied.

LLM REQUIREMENT: The language model must be part of the operator-note interpretation path. Using an LLM only for `plan_summary`, `documentation`, or `cosmetic text` does not satisfy this requirement.

03. End-to-End Processing Flow

The LLM understands human language; deterministic code validates the interpretation; the optimizer performs the mathematical scheduling.



CORE IDEA: Human notes are not directly trusted as math. They are first converted to a fixed structured format, checked by guardrails, and only then applied to the optimization model.

04. Operator Notes & Supported Directives

Each scenario contains 1-3 operator notes. Some notes affect the current schedule; others are realistic distractors and must be ignored. Hidden cases may express the same directive using different wording.

4.1 Supported directive types

Directive type	Meaning	Required structured_adjustment
solar_reduction	Reduce usable solar during specific hours.	{"hours": [...], "factor": number}
minimum_battery_reserve	Keep battery energy at or above a required level.	{"hours": [...], "minimum_energy_kwh": number}
no_charge_window	Battery charging is unavailable during specific hours.	{"hours": [...]}
no_discharge_window	Battery discharging is unavailable during specific hours.	{"hours": [...]}
max_grid_window	Grid import may not exceed a stated amount during specific hours.	{"hours": [...], "max_grid_kwh": number}
no_op	The note does not affect the current 24-hour energy schedule.	null

4.2 Simple examples

Operator note	Expected interpretation
"Solar output will drop to about 20% from 1 PM to 3 PM."	solar_reduction; hours [13,14]; factor 0.2
"Do not charge the battery between 2 PM and 4 PM."	no_charge_window; hours [14,15]
"Keep at least 120 kWh in reserve from 6 PM until 9 PM."	minimum_battery_reserve; hours [18,19,20]; 120 kWh
"The cafeteria menu changes tomorrow."	no_op

05. Clauses & Optimization Task

5.1 Operator-note clauses

- Every operator note must produce exactly one `directive_interpretation` entry. Entries must be returned in `note_index` order: 0, 1, ... N-1.
- Only the directive types listed in Section 04 are accepted.
- Relevant notes must be applied to the optimization before scheduling.
- Irrelevant notes must use `applies = false`, `directive_type = no_op`, and `structured_adjustment = null`.
- The same directive may appear in different wording in hidden cases.
- Time windows use whole-hour intervals. The start hour is included and the end hour is excluded: 1 PM to 3 PM means hours [13,14].

- The LLM must not invent demand, solar, tariff, battery limits, or unsupported directive types.
- A schedule that interprets a note correctly but does not apply it is still incorrect.
- For every non-no_op directive, applies must be true. no_op is the only directive allowed with applies = false.
- Every hours array inside structured_adjustment must contain unique integers from 0 through 23 in ascending order.
- For solar_reduction, factor means the usable fraction that remains. Example: an 80% reduction means factor = 0.2.
- Organizer valid scoring scenarios are feasible and will not require mutually contradictory hard directives to be satisfied at the same time.

5.2 Optimization objective

After applying all valid directive adjustments, minimize the total cost of grid electricity over the 24-hour horizon:

$$\text{total_cost_bdt} = \text{SUM}(\text{grid_kwh}[h] * \text{tariff_bdt_per_kwh}[h]) \text{ for } h = 0..23$$

Lower cost is better, but a low-cost schedule is invalid if it breaks any energy, battery, or operator-directive rule.

5.3 How directives change the math

Directive	Deterministic effect used by the optimizer
solar_reduction	effective_solar[h] = original_solar[h] * factor for each listed hour.
minimum_battery_reserve	battery_energy_after_kwh[h] >= max(base minimum_energy_kwh, directive minimum_energy_kwh) for each listed hour.
no_charge_window	battery charge amount = 0 in the listed hours.
no_discharge_window	battery discharge amount = 0 in the listed hours.
max_grid_window	grid_kwh[h] <= max_grid_kwh in the listed hours.
no_op	No change to the optimization model.

06. API Contract

The judge harness exercises only the endpoints below. Endpoint names must match exactly.

Endpoint	Requirement
GET /health	Return HTTP 200 with a JSON object containing status = "ok" when the service is ready.
POST /optimize-energy	Accept one scenario JSON object and return one interpretation + optimization-plan JSON object.

6.1 HTTP response codes

Code	Meaning
200	Successful health response or successful optimization response.
400	Malformed JSON or structurally invalid request.
422	Optional: semantically invalid but well-formed request.
500	Controlled internal error. Do not expose secrets or raw stack traces.

6.2 Health response

```
{
  "status": "ok"
}
```

07. Request Schema

POST /optimize-energy accepts one JSON object. The hours array must contain exactly 24 entries for hours 0 through 23. operator_notes must contain 1-3 non-empty natural-language strings, each referring to the same 24-hour scenario.

7.1 Top-level fields

Field	Type	Requirement
scenario_id	string	Unique synthetic scenario identifier.
operator_notes	array[1..3] of string	Natural-language campus operator notes to interpret.
hours	array[24]	Hourly demand, solar availability, and grid tariff.
battery	object	Battery capacity, starting energy, reserve, and hourly limits.

7.2 Hour entry

Field	Type	Meaning
hour	integer	Unique integer from 0 to 23.
demand_kwh	number	Campus demand that must be supplied in this hour.
solar_kwh	number	Base solar energy available before operator-note adjustments.
tariff_bdt_per_kwh	number	Grid electricity price for this hour.

7.3 Battery object

Field	Meaning
capacity_kwh	Maximum energy the battery can store.
initial_energy_kwh	Battery energy at the start of hour 0.
minimum_energy_kwh	Base reserve level the battery must never go below.
max_charge_kwh_per_hour	Maximum energy that may be added in one hour.
max_discharge_kwh_per_hour	Maximum energy that may be removed in one hour.

7.4 Example request shape

```
{
  "scenario_id": "GRID-101",
  "operator_notes": [
    "Solar output will drop to about 20% from 1 PM to 3 PM.",
    "Do not charge the battery between 2 PM and 4 PM.",
    "The cafeteria menu changes tomorrow."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 180, "solar_kwh": 0, "tariff_bdt_per_kwh": 7},
    ... 22 more hourly entries ...,
    {"hour": 23, "demand_kwh": 200, "solar_kwh": 0, "tariff_bdt_per_kwh": 9}
  ],
  "battery": {
    "capacity_kwh": 500,
    "initial_energy_kwh": 200,
    "minimum_energy_kwh": 50,
    "max_charge_kwh_per_hour": 100,
    "max_discharge_kwh_per_hour": 100
  }
}
```

08. LLM Interpretation Guardrails

LLM output must be treated as untrusted structured data until deterministic validation passes.

Guardrail	Requirement
Allowed types	directive_type must be one of the values in Section 04.

Guardrail	Requirement
Note mapping	note_index must identify an existing operator note and each note must appear once.
Hours	Every listed hour must be a unique integer from 0 through 23, returned in ascending order.
Solar factor	For solar_reduction, factor must be between 0 and 1 inclusive.
Battery reserve	Reserve values must be finite, non-negative, and not exceed battery capacity.
Grid cap	max_grid_kwh must be finite and non-negative.
No invention	The interpretation may not change base demand, tariff, or battery parameters unless a supported directive explicitly allows it.
Final replay	The completed schedule is replayed after optimization to verify every extracted directive was actually followed.
applies semantics	For no_op: applies = false and structured_adjustment = null. For every other directive: applies = true and structured_adjustment must match the required shape in Section 04.
Feasible judge scenarios	Valid organizer scoring scenarios will have a feasible ground-truth interpretation and will not require contradictory hard directives.

SAFE FAILURE: If the LLM returns malformed or unsupported structured output, the service must handle it in a controlled way. The service must not silently invent a new directive type or crash.

09. Battery & Energy Rules

The existing GridWise energy rules remain unchanged. The judge independently replays the final schedule hour by hour using the effective solar and any additional operator directives.

9.1 Battery state

charge: $E_{\text{after}} = E_{\text{before}} + \text{battery_kwh}$
 discharge: $E_{\text{after}} = E_{\text{before}} - \text{battery_kwh}$
 idle: $E_{\text{after}} = E_{\text{before}}$ and $\text{battery_kwh} = 0$

9.2 Battery bounds

$\text{minimum_energy_kwh} \leq E_{\text{after}} \leq \text{capacity_kwh}$

If a minimum_battery_reserve directive is active, its reserve may be higher than the base minimum for those hours.

9.3 Hourly charge/discharge limits

if action = charge: $\text{battery_kwh} \leq \text{max_charge_kwh_per_hour}$
 if action = discharge: $\text{battery_kwh} \leq \text{max_discharge_kwh_per_hour}$

9.4 Solar usage

$0 \leq \text{solar_used_kwh} \leq \text{effective_solar_kwh}$ for that hour

Unused solar is curtailed. Grid export is not part of this challenge.

9.5 Energy balance

$\text{grid_kwh} + \text{solar_used_kwh} + \text{battery_discharge_kwh}$
 $=$
 $\text{demand_kwh} + \text{battery_charge_kwh}$

9.6 End-of-day battery neutrality

$\text{final_battery_energy_after_kwh} = \text{initial_energy_kwh}$

WHY THIS RULE EXISTS: The starting battery may shift energy between hours, but it cannot be consumed as a free one-time source by ending the day at a lower state of charge.

10. Response Schema

A successful POST /optimize-energy response must include both the operator-note interpretation and the final 24-hour schedule.

10.1 Top-level response fields

Field	Type	Requirement
scenario_id	string	Must match the request scenario_id.
directive_interpretation	array	One machine-checkable interpretation entry for every operator note.
hourly_plan	array[24]	One plan entry for every hour 0 through 23.
total_grid_kwh	number	Sum of grid_kwh across all 24 hours.
total_cost_bdt	number	Calculated total grid electricity cost.
peak_grid_kwh	number	Maximum hourly grid_kwh in the returned plan.
plan_summary	string	Short human-readable explanation of the final strategy.

10.2 Directive interpretation entry

Field	Requirement
note_index	Zero-based index of the corresponding operator_notes entry.
applies	true for every applicable non-no_op directive; false only for no_op.
directive_type	One supported directive type from Section 04. no_op is required when applies = false.
structured_adjustment	Exact machine-checkable object required by Section 04, or null only for no_op.
explanation	Short explanation of the interpretation.

10.3 Hourly plan entry

Field	Allowed value / meaning
hour	Integer 0 through 23.
grid_kwh	Non-negative grid energy purchased in this hour.
solar_used_kwh	Solar energy used in this hour; cannot exceed effective solar.
battery_action	Exactly one of: charge, discharge, idle.
battery_kwh	Non-negative magnitude of the battery action. Must be 0 when idle.
battery_energy_after_kwh	Battery energy immediately after completing this hour.

10.4 Example interpretation fragment

```
"directive_interpretation": [
  {
    "note_index": 0,
    "applies": true,
    "directive_type": "solar_reduction",
    "structured_adjustment": {"hours": [13, 14], "factor": 0.2},
    "explanation": "Solar availability is reduced during panel cleaning."
  },
  {
    "note_index": 2,
    "applies": false,
    "directive_type": "no_op",
    "structured_adjustment": null,
    "explanation": "This note does not affect today's energy schedule."
  }
]
```

11. Exact Validation & Hidden Evaluation

Hidden evaluation checks both language understanding and energy optimization. Teams should not assume that only `total_cost_bdt` or the free-text explanation is checked.

11.1 Interpretation checks

- Correctly identify whether each note applies or is `no_op`.
- Return the correct directive type.
- Extract the correct hours and numeric values within the allowed tolerance.
- Remain robust when the same directive is paraphrased in different ways.
- Return one entry per note in `note_index` order, with no missing or duplicate mappings.
- Match the required `structured_adjustment` shape for the selected directive type.
- Use `applies = false` only for `no_op`; all applicable directives use `applies = true`.

11.2 Downstream application checks

- The judge recomputes effective solar after `solar_reduction` directives.
- The judge verifies reserve, no-charge, no-discharge, and grid-cap directives directly against `hourly_plan`.
- Correct extraction without correct downstream application does not pass the case.

11.3 Existing GridWise consistency checks

- `hourly_plan` contains exactly 24 unique hours, 0 through 23.
- Required numeric values are finite and non-negative.
- Battery transitions, capacity, minimum energy, and hourly rate limits are valid.
- Solar usage never exceeds effective available solar.
- The energy-balance equation holds every hour.
- Final battery energy equals initial battery energy.
- `total_grid_kwh`, `total_cost_bdt`, and `peak_grid_kwh` match values recalculated from `hourly_plan`.

11.4 Hidden language variation

The same underlying rule may be written differently in hidden cases. For example, all three notes below mean the same solar-reduction directive:

Hidden scoring notes will map to exactly one supported directive type or no_op. They may be paraphrased, but they will not require an unpublished directive type or an impossible combination of hard constraints.

Example wording
"PV production will drop to about 20% between 13:00 and 15:00."
"Panel washing from one until three will leave roughly one-fifth of normal solar output."
"Expect an 80% reduction in rooftop solar during the 1-3 PM maintenance window."

NO BYTE-FOR-BYTE MATCHING: Equivalent valid optimal schedules may differ. The judge evaluates structured interpretation, directive application, schedule validity, and recalculated cost rather than exact JSON equality with one reference plan.

11.5 Numeric tolerance

For normal floating-point arithmetic, judge calculations should treat values within an absolute tolerance of 0.01 kWh or 0.01 BDT as equivalent unless the official judge package specifies a stricter value.

12. Canonical Specification Note

For this challenge, this Problem Statement is the canonical source for endpoint names, request/response fields, operator-note directive types, interpretation guardrails, battery behavior, energy accounting, and optimization validity rules.

The separate Participant Guide & Evaluation Rubric remains canonical for deployment, repository policy, submission procedure, performance requirements, evaluation weights, penalties, and tie-breakers.

FINAL REMINDER: Understand the operator notes first. Validate the extracted directives. Apply them to the optimization. Then produce a valid 24-hour schedule and minimize cost.